

Predicting Extreme Climate Behavior with Machine Learning

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Today's goals

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Module 1

Unsupervised Learning: Dimensionality Reduction

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Course Introduction

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Course Updates and Accessibility Support

Reading • 1 min

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Earn Academic Credit for your Work!

Reading • 10 min

✓

Course Support

Reading • 10 min

●

Introduction to the Course

Video • 4 min

●

Meet the Instructor

Video • 1 min

✓

Introduction to Unsupervised Learning and Techniques

Video • 5 min

●

Unsupervised Learning and Climate Anomalies

Discussion Prompt • 30 min

Principal Component Analysis

✓

PCA Overview

Video • 18 min

✓

PCA in Terms of SVD

Video • 23 min

●

Principal Component Analysis for Extremes and Application to U.S. Precipitation

Reading • 30 min

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Principal Component Analysis and Singular Value Decomposition (SVD)

Graded Assignment • 20 min

Module Assignment

●

PCA on Soil Temperature Data: Notebook Walkthrough

Video • 7 min

●

PCA on Soil Temperature Data: Notebook Walkthrough

Lab • 30 min

●

PCA on Soil Moisture Data

Programming Assignment • 1h

Module 2

Unsupervised Learning: Clustering

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Module 3

Supervised Learning: Regressions

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Module 4

Supervised Learning: Logistic Regression, Decision Trees, and SVMs

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Module 5

Supervised Learning: Neural Networks

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Principal Component Analysis for Extremes and Application to U.S. Precipitation

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The next discussion prompt will be based on the reading [“Principal Component Analysis for Extremes and Application to U.S. Precipitation”](#) (external link; opens in a new tab).

From the paper's abstract, this research paper delves into analyzing extremal behavior through the lens of a most efficient basis of vectors. The method is analogous to PCA but is based on methods from **extreme value analysis**. Specifically, rather than decomposing a covariance or correlation matrix, they obtain basis vectors by performing an eigen decomposition (see video on PCA/SVD) of a matrix that describes pairwise extremal dependence. The researchers then apply the method to precipitation observations over the contiguous United States. They then discover that the time series of large coefficients associated with the leading eigenvector shows very strong evidence of a positive trend, and there is evidence that large coefficients of other eigenvectors have relationships with *El Niño–Southern Oscillation*. Though I do not expect you to know every single mathematical deviation/nuance that is calculated in this paper, this paper gets you into the mode of reading novel published research in this pertinent field of climate analysis. After having read the paper, please answer each of the questions to the best of your ability. There is no specific word limit, but thoughtful and insightful responses will be beneficial to you and your peers.

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